

LIFESTYLE FEATURE DESIGN PLAN

Offline-first community health screening desktop application

Authoritative product, UX, data, security, and implementation reference

Version: 1.0 **Date:** 12 August 2026

Status: Approved design direction; implementation must be delivered through separately reviewed tasks.

Core principle

Show only the information required for the patient's current path. Keep baseline history separate from the weekly encounter record.

1. Purpose and boundaries

This document defines the Lifestyle feature that follows Vitals in the screening workflow. It is the reference Luna must read before implementation prompts are executed. The feature collects analyzable alcohol, tobacco/nicotine, physical-activity, work, and other-activity information while remaining fast, respectful, offline capable, and non-diagnostic.

Objectives

- Collect quantitative weekly data for alcohol, tobacco/nicotine, and physical activity.
- Collect structured plus descriptive occupation and other-activity context.
- Minimize visible questions through conditional branching and collapsible cards.
- Create a stable baseline profile and immutable baseline history without rewriting earlier encounters.
- Support Save Draft, recovery, and Continue using the application's established security and transactional boundaries.
- Prepare clean domain data for later reporting and FHIR mapping without implementing either in this feature.

Out of scope

- Diagnosis of alcohol-use disorder, nicotine dependence, or physical inactivity.
- Automated clinical recommendations, referrals, printing, reporting, remote sync, or FHIR persistence.
- Changes to authentication, Patient Search, session creation, encounter creation, or Vitals behavior.
- Food and over-the-counter medication questionnaires; they are separate later modules.

Clinical posture

The module is a screening and data-collection tool. It may calculate transparent totals and validated questionnaire scores when separately approved, but it must not attach stigmatizing labels or substitute for clinical assessment.

2. Product model: baseline plus weekly screening

Lifestyle uses two related but distinct records. A baseline describes usual or historical status. A weekly record describes only the selected screening period, normally the past seven days.

Dimension	Baseline profile	Weekly encounter record
Purpose	Usual/history status and branching context.	What occurred during this screening period.
Frequency	Initial completion; update only when circumstances change.	Created for every Lifestyle screening.
UI entry	A compact Baseline button opens an inline panel.	The primary questions shown in each card.
History	Versioned; previous versions remain immutable.	Immutable after encounter finalization under lifecycle policy.
Relationship	Active version helps select weekly questions.	References the baseline version used when the encounter was completed.
Unknowns	Unknown and Prefer not to answer remain explicit.	Blank, zero, unknown, and declined are distinct.

Baseline versioning rule

Updating a baseline creates a new version with an effective timestamp and actor attribution. It must never silently overwrite the prior version. Existing weekly records retain their original baseline-version reference. General audit metadata must describe the action without copying clinical answers.

Required status values

Domain	Statuses
Alcohol	CURRENT, FORMER, NEVER, UNKNOWN, DECLINED
Tobacco/nicotine	CURRENT_DAILY, CURRENT_SOME_DAYS, FORMER, NEVER, UNKNOWN, DECLINED
Work	EMPLOYED, SELF_EMPLOYED, FARMING, STUDENT, HOMEMAKER_CAREGIVER, UNEMPLOYED, RETIRED, UNABLE_TO_WORK, OTHER, DECLINED

3. Overall user experience

The Lifestyle workspace presents five cards in order. Only the active card expands. Completed cards collapse automatically but remain reopenable until the encounter becomes non-editable.

Order	Card	Default compact summary
1	Alcohol	Baseline status + weekly completion/result
2	Tobacco and nicotine	Baseline status + weekly completion/result
3	Physical activity	Weekly minutes/status
4	Work and occupation	Work baseline + weekly variation
5	Other activity	Weekly completion/status

Card states

State	Indicator	Behavior
Not started	Neutral circle	Compact; next required card may open.
In progress	Blue accent	Expanded and editable.
Complete	Green check	Collapsed; summary visible.
Baseline review required	Amber alert	Weekly use conflicts with baseline; review required.
Validation error	Red marker	Remain open; focus first invalid field.
Locked	Lock state	Readable but not editable under encounter lifecycle.

Primary controls

- Save Draft persists incomplete valid work and remains in Lifestyle.
- Continue requires every required card to be complete, persists successfully, then opens the next approved module.
- Baseline buttons expand/collapse inline and never navigate away from the encounter.
- Back or card reopening does not discard unsaved edits; explicit discard confirmation is required when applicable.
- Keyboard order follows the visual path, Enter must not accidentally advance multi-step sections, and all controls require visible focus.

Example compact view

✓ **Alcohol** No use this week
✓ **Tobacco** No use this week
✓ **Physical Activity** 150 minutes
■ **Work Activity** Not completed
■ **Other Activity** Not started

4. Alcohol design

Alcohol uses a baseline status to shorten the weekly path. Questions use neutral language and locally appropriate drink examples. The data model stores standardized counts separately from display guidance.

Alcohol baseline panel

Field	Type / options	Rule
Ever consumed alcohol	YES / NO / UNKNOWN / DECLINED	Required for baseline completion.
Consumed in past 12 months	YES / NO / UNKNOWN / DECLINED	Shown when ever use is Yes or Unknown.
Current status	CURRENT / FORMER / NEVER / UNKNOWN / DECLINED	Derived only through approved explicit rules; user can confirm.
Common beverage types	Beer, wine, spirits, palm wine, traditional/home-brewed, other	Multi-select; shown when applicable.
Other beverage description	Short text	Required only when Other selected.
Validated baseline instrument	Deferred configuration	AUDIT-C may be added only through a separately approved clinical task.

Weekly branching

Baseline status	First weekly question	If No	If Yes
Current	Did you consume alcohol during the past 7 days?	Record 0 drinking days; complete.	Open quantitative fields.
Former / Never	Did you consume any alcohol during the past 7 days?	Preserve baseline; complete.	Open quantitative fields and require baseline review.
Unknown / Declined	Can alcohol use be recorded for the past 7 days?	Record Unknown/Declined, not zero.	Open quantitative fields; offer baseline review.
Missing	Complete Alcohol Baseline.	Not applicable.	Baseline must be saved before weekly completion.

Weekly quantitative fields after Yes

Field	Type	Validation
Drinking days	Integer	1-7
Total standardized drinks	Decimal or integer per approved convention	Greater than 0; upper safety limit triggers validation, not diagnosis.
Largest number in one day	Decimal/integer	Greater than 0 and not greater than total.
Days at largest amount	Integer	1 through drinking days.
Beverage types	Multi-select	At least one if configured required.
Clarifying note	Short optional text	Length-limited; never placed in audit metadata.

Conflict behavior

A Former or Never baseline combined with weekly Yes creates an amber “Review baseline” state. Weekly answers remain intact while the baseline panel opens. The baseline must be confirmed or versioned before Continue; Save Draft remains allowed. The UI must not automatically reclassify the patient without confirmation.

5. Tobacco and nicotine design

Tobacco includes combustible, smokeless, and nicotine products. A zero-use week does not automatically change a current user to former status.

Tobacco baseline panel

Field	Type / options	Rule
Ever regularly used	YES / NO / UNKNOWN / DECLINED	Required.
Current use frequency	Every day / some days / not at all / unknown / declined	Shown when applicable.
Current status	Current daily / current some days / former / never / unknown / declined	Confirmed baseline value.
Products	Cigarettes, rolled tobacco, cigar/pipe, smokeless/chewing, snuff, shisha/hookah, vape/e-cigarette, other	Multi-select.
Former-use stop date	Approximate month/year or year	Optional; do not invent precision.
Other product description	Short text	Required when Other selected.

Weekly branching

Baseline	Question	No	Yes
Current	Did you use any tobacco or nicotine product in the past 7 days?	Record zero use for week; keep baseline Current.	Open product rows.
Former / Never	Did you use any tobacco or nicotine product in the past 7 days?	Preserve baseline; complete.	Open product rows; require baseline review.
Unknown / Declined	Can tobacco/nicotine use be recorded for the past 7 days?	Record explicit response, not zero.	Open product rows; offer baseline review.

Product-row fields

Field	Rule
Product type	One controlled product type per row; no duplicates.
Days used	Integer 1-7.
Average quantity per use day	Positive numeric value.
Unit	Sticks/cigarettes, sessions, portions/pinches, pods/cartridges, or Other.
Other unit	Short text required only for Other.
Secondhand smoke exposure	No / Yes / Unknown / Declined; optional by approved scope.

6. Physical activity design

Use “Physical Activity,” not only “Exercise,” because farming, domestic work, construction, and walking may be the patient’s primary activity. The weekly record separates domains so totals remain interpretable.

Weekly entry question

During the past 7 days, did you perform physical activity for at least 10 continuous minutes?

Options: Yes / No / Unable to answer / Prefer not to answer. The 10-minute threshold must remain configurable if the approved protocol changes.

If Yes: select domains

- Work or farming
- Walking or cycling for transportation
- Household or caregiving activity
- Exercise, sport, or recreation

Fields per selected domain/activity row

Field	Type	Rule
Activity category	Controlled value	Must belong to selected domain.
Description	Short optional text	Useful for local activities.
Intensity	Light / moderate / vigorous	Display plain-language examples.
Days in past 7 days	Integer	1-7.
Average minutes per active day	Integer	Positive, realistic configured maximum.
Weekly minutes	Calculated	Days x minutes; retain inputs.

Sedentary time

Collect usual sitting or reclining time per day, excluding sleep, using hours and minutes or total minutes. Unknown and declined remain distinct. Do not subtract sedentary time from activity totals.

Calculation policy

- Store raw days, minutes, and intensity, then calculate totals deterministically.
- Retain moderate and vigorous minutes separately; do not silently convert them unless a documented derived metric is added.
- Prevent double entry by reminding the screener not to repeat an activity already captured in another domain.
- Calculated totals support display and later analysis; they do not create a diagnosis.

7. Work and other activity

Occupation is baseline data with a very short weekly variance question. Other activity is weekly and captures relevant activity not already recorded.

Work baseline

Field	Options / type
Employment status	Employed, self-employed, farming, student, homemaker/caregiver, unemployed, retired, unable to work, other, declined
Occupation or job title	Short free text
Usual physical demand	Mostly sitting, standing, walking, moderate labor, heavy labor, varies, unknown
Typical workdays per week	Integer 0-7
Typical hours per workday	Decimal or hours/minutes
Shift pattern	Day, evening, night, rotating, irregular, not applicable
Description	Optional length-limited text

Weekly work question

Did you perform your usual work during the past 7 days?

Yes - usual amount / Yes - less than usual / Yes - more than usual / No / Not applicable / Unknown / Prefer not to answer. If the occupation changed, the screener opens Work Baseline and creates a new version.

Other activity

Ask: "Did you perform any other regular physical activity during the past 7 days that was not already recorded?" If Yes, capture category, description, days, minutes per day, and intensity. Categories include farming/gardening, household work, caregiving, religious/community activity, walking for errands, sport, and Other.

Duplication guard

The application may warn about similar descriptions or identical category/day/minute combinations, but must not delete or merge entries automatically. The screener decides whether the activity is a duplicate.

8. Data architecture

The exact physical schema must follow the repository's established migration, identifier, timestamp, and repository conventions. The following is a logical model, not permission to bypass architecture boundaries.

Entity	Purpose	Key relationships
Lifestyle baseline	Stable patient-level baseline container.	Patient, installation; one active logical baseline.
Baseline version	Immutable domain snapshot effective from a timestamp.	Baseline, actor; alcohol/tobacco/work version data.
Lifestyle draft	Encounter-owned parent for weekly answers.	Patient, encounter, historical session/location attribution, version.
Alcohol weekly	Weekly alcohol response and quantitative totals.	One per Lifestyle draft.
Tobacco weekly	Weekly tobacco summary.	One per Lifestyle draft.
Tobacco product row	Stable-ID product quantities.	Many per tobacco weekly record.
Physical activity weekly	Top-level response and sedentary time.	One per Lifestyle draft.
Activity row	Stable-ID activity-domain inputs.	Many per physical activity weekly record.
Work weekly	Weekly variance from baseline.	One per Lifestyle draft.
Other activity row	Stable-ID uncaptured activities.	Many per Lifestyle draft.

Data invariants

- One Lifestyle draft per encounter; patient, encounter, session, location, and installation attribution is authoritative and immutable.
- Renderer never supplies actor, version authority, timestamps, baseline history authority, calculated scores, session, location, or installation.
- Stable child IDs support add/update/remove reconciliation; cross-draft and duplicate IDs are rejected.
- Blank, zero, No, Unknown, Not applicable, and Prefer not to answer are distinct states.
- Existing created timestamps are preserved; unchanged child rows must not receive artificial updated timestamps.
- A completed weekly record stores the baseline-version references used for interpretation.

Suggested controlled codes

Use internal enums or controlled codes rather than UI labels for status, response, product, unit, activity domain, intensity, employment status, and shift pattern. Labels may later be localized without changing stored meaning.

9. Validation and workflow rules

Rule area	Save Draft	Continue
Incomplete cards	Allowed when entered fields are individually valid.	Rejected with focus on first incomplete required card.
Missing baseline	Draft can be saved.	Required baseline domains must be complete.
Former/Never + weekly Yes	Save answers and amber review state.	Baseline review/confirmation required.
Quantitative dependency	Partial values allowed if non-contradictory.	Every field required by Yes must be complete.
No response	Store No and approved zeros.	Allowed when logically complete.
Unknown/Declined	Store explicit code, never coerce to zero.	Allowed according to field policy.
Persistence	Transactional save with optimistic concurrency.	Persist first; navigate only after commit.

Cross-field validation examples

- Alcohol drinking days cannot be zero when weekly use is Yes.
- Largest one-day alcohol amount cannot exceed total weekly drinks.
- Days at largest amount cannot exceed drinking days.
- A tobacco product row requires product, days, quantity, and unit.
- Activity days and minutes must be positive when an activity row exists.
- No weekly use must not coexist with positive quantities.
- Former/Never conflict is a review requirement, not an automatic clinical conclusion.
- Text inputs are trimmed, normalized under existing conventions, and length-limited.

Recovery and lifecycle

Lifestyle must follow the already approved recovery pattern: an existing editable draft may be restored after logout/login, restart, or date rollover without rewriting historical attribution. First-draft creation must use the current encounter/session authority defined by the established workflow. Closed or non-editable encounters remain blocked.

10. Offline-first behavior and transactions

All feature behavior must work without network access. The local database is the immediate system of record for the desktop workflow. This design does not add remote sync.

Save transaction

- Validate authenticated actor, permission, trusted IPC sender, installation, patient, encounter ownership, and lifecycle.
- Load authoritative current draft, baseline versions, child records, and optimistic version.
- Validate baseline/weekly branching and normalized inputs.
- Insert or version approved baseline changes without overwriting earlier versions.
- Reconcile weekly child rows by stable ID, skipping updates for unchanged rows.
- Update the parent draft/version under the established clock.
- Write approved sanitized audit and outbox records.
- Return success only after commit; any failure rolls back every change.

Concurrency

A stale renderer version returns the established sanitized conflict result. No automatic last-write-wins behavior is permitted. Baseline updates and weekly saves that depend on them must be coordinated transactionally or through an explicitly designed version contract.

Future sync preparation

Use stable identifiers, version numbers, created/updated timestamps, tombstone or approved deletion semantics, and immutable encounter attribution consistent with the project architecture. Do not create network calls, remote identifiers, merge algorithms, or FHIR resources in this task.

11. Security, privacy, and accessibility

Security boundaries

- All authorization and persistence remain in the main process; renderer access is through narrow validated IPC contracts.
- Renderer input cannot assert actor, role, patient ownership, installation, location, session, encounter lifecycle, timestamps, audit metadata, baseline status history, or calculated authority.
- Do not log clinical answers, patient identifiers, notes, raw database errors, SQL, or full IPC payloads.
- Return stable sanitized result codes and user-safe messages.
- Do not introduce bypass, force, recovery, admin, or override flags.

Privacy-sensitive interaction

- Use neutral wording and a visible “Prefer not to answer” option where approved.
- Do not display stigmatizing labels such as alcoholic, addict, or inactive.
- Keep baseline summaries concise so sensitive detail is not unnecessarily exposed on screen.
- Do not prefill a new week as if prior behavior occurred again; prior answers may be shown only as clearly labeled reference.

Accessibility

- Every input has a programmatic label; required state is not conveyed by color alone.
- Expandable panels expose expanded/collapsed state to assistive technology.
- Validation summaries link to fields; focus moves predictably.
- Touch/click targets, contrast, zoom, and supported desktop window sizes follow existing UI standards.
- Quantities include units in labels and errors; examples do not replace labels.

12. Testing strategy

Unit and domain tests

- Every baseline status and weekly Yes/No/Unknown/Declined branch.
- Alcohol and activity calculations, boundaries, and cross-field contradictions.
- Tobacco product and activity stable-ID reconciliation, duplicates, removal, reordering, and cross-draft rejection.
- Baseline version creation, prior-version preservation, and weekly version references.
- Unchanged-row timestamp preservation and deterministic injected-clock behavior.

Application and persistence integration

- First draft, update, Save Draft, Continue, optimistic conflict, and rollback.
- Actor/permission/trusted-sender, patient/encounter/installation ownership, and lifecycle rejection.
- Existing-draft recovery across logout/login, restart, and deployment-local date rollover.
- No partial baseline or weekly persistence after a failure.
- Audit/outbox contains no clinical values and only emits success after commit.

Renderer and UX tests

- Only the required active fields are visible; irrelevant branches are absent.
- Baseline button expands/collapses inline; successful save closes and updates summary.
- Current, former, never, missing, unknown, and declined paths render correctly.
- Former/Never + weekly Yes preserves weekly answers and requires baseline review.
- Completed cards collapse, remain reopenable, and show accessible state.
- Save Draft does not navigate; Continue persists before navigation.
- Keyboard, focus, error announcement, required indicators, and responsive desktop layout.

Regression gates

Run repository formatting, lint, type checking, complete tests, verification, build, migration checks, diff checks, and a safe desktop smoke test. P0-P4, authentication, Patient Search, sessions, encounters, Vitals, audit, outbox, and recovery behavior must remain passing.

13. Implementation sequence for Luna prompts

Do not ask Luna to implement the entire feature at once. Each task must begin from an exact reviewed main baseline, create one focused branch, read this PDF and repository instructions, run focused and full gates, push, stop, and wait for review before merge.

Task	Scope	Exit condition
L1 - Domain decisions	Confirm repository patterns, field dictionary, enums, branching state machine, and migration plan. No implementation if unresolved.	Approved design addendum or no-change confirmation.
L2 - Persistence foundation	Migrations, repositories, baseline versioning, weekly parent and child models, constraints.	Persistence/security tests pass; no UI.
L3 - Main-process service	Authorization, trusted IPC, load/save, concurrency, transaction, audit/outbox.	Service and integration tests pass.
L4 - Alcohol UI	Progressive card, baseline panel, weekly branching and validation.	Alcohol UX and integration tests pass.
L5 - Tobacco UI	Baseline, weekly product rows, conflict review.	Tobacco tests pass.
L6 - Activity UI	Physical activity, work baseline/variance, other activities, calculations.	Activity tests pass.
L7 - Workflow integration	Card progress, Save Draft, recovery, Continue to next approved module boundary.	End-to-end Lifestyle workflow passes.
L8 - Hardening	Accessibility, responsive layout, failure/recovery, proof searches, final regression.	All quality gates and smoke tests pass.

Task discipline

- No task may silently broaden into recommendations, Food, OTC, sync, FHIR, reports, or printing.
- A migration must be additive, reversible according to project policy, and tested against fresh and upgraded databases.
- After every branch is pushed, perform an independent review before merging.
- If repository architecture conflicts with this logical plan, Luna must stop and report the exact conflict rather than invent a parallel pattern.

14. Acceptance criteria

Area	The feature is acceptable when...
Simplicity	Only path-relevant questions appear; five-card workflow is clear and fast.
Baseline	Alcohol, tobacco, and work baselines open inline, close after save, and preserve version history.
Weekly data	Alcohol, tobacco, and activity values are quantitative and bound to the correct encounter period.
Status conflict	Former/Never + weekly Yes requires review without discarding weekly answers or auto-reclassifying.
Correct semantics	Zero, No, blank, Unknown, Not applicable, and Declined remain distinct.
Navigation	Save Draft stays; Continue requires completion, commits, then advances.
Recovery	Editable drafts survive logout/login, restart, and date rollover with historical attribution intact.
Security	Main-process authority, trusted IPC, ownership, lifecycle, sanitization, and privacy rules hold.
Offline	Full feature works without internet; no remote dependency is introduced.
Quality	Focused, integration, renderer, regression, migration, build, and visual checks pass.

Design decisions requiring later explicit approval

- Exact standard-drink conversion guidance for Cameroon and local beverage/container examples.
- Whether AUDIT-C is included at baseline and how validated scoring is presented.
- Exact physical-activity duration threshold and any derived guideline comparison.
- Whether secondhand-smoke exposure is included in the first Lifestyle release.
- The next module opened by Continue after Lifestyle.
- FHIR resource mapping, recommendations, referral thresholds, and reporting.

Approval statement

This plan defines the intended Lifestyle design. Implementation prompts may refine technical names to match the repository, but may not change the product behavior, authority boundaries, history rules, or scope without explicit approval.

15. References

These references support the questionnaire structure. They do not automatically authorize every instrument, threshold, or scoring rule for implementation.

Reference	Design relevance
World Health Organization - Alcohol Use Disorders Identification Test (AUDIT)	Validated alcohol screening foundation and primary-care guidance. https://www.who.int/publications/i/item/WHO-MSD-MSB-01.6a
WHO - Global Physical Activity Questionnaire (GPAQ)	Separates work, transport, and recreation; collects frequency, duration, intensity, and sedentary behavior. https://www.who.int/publications/m/item/global-physical-activity-questionnaire
WHO - GPAQ Analysis Guide	Calculation and data-quality context. https://cdn.who.int/media/docs/default-source/ncds/ncd-surveillance/gpaq-analysis-guide.pdf
NIAAA - Screen and Assess	AUDIT-C-style frequency, typical amount, and high-quantity occasion questions. https://www.niaaa.nih.gov/health-professionals-communities/core-resource-on-alcohol/screen-and-assess-use-quick-effective-methods
NIAAA - What Is a Standard Drink?	Need to explain standardized drink quantities rather than only containers. https://www.niaaa.nih.gov/alcohols-effects-health/what-standard-drink
CDC/NCHS - Tobacco use definitions	Current every-day/some-days/not-at-all classification context. https://www.cdc.gov/nchs/hs/sources-definitions/tobacco-use.htm

End of document